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EE1030/EE2100: Matrix Theory

Lecture 22: Matrices and Linear Transformations

Topics :

1. Linear Transformation - Introduction
 2. Determining Matrix Corresponding to a Linear Transformation
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Every Linear Transformation can be represented by a Matrix

Let $\vec{f}(\vec{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ & $\vec{f}(\vec{x})$ is linear. $\vec{f}(\vec{x})$ can be represented as $T\vec{x}$
 $\downarrow$ linear transformation $\downarrow$ Transformation matrix $T \in \mathbb{R}^{m \times n}$

Proof :- $\vec{x} \in \mathbb{R}^n = \sum_{i=1}^n x_i \vec{e}_i \quad \vec{e}_i \in \mathbb{R}^n$

$$\begin{aligned} \because \vec{f}(\vec{x}) \text{ is linear} &\Rightarrow \vec{f}(\vec{x}) = \vec{f}\left(\sum_{i=1}^n x_i \vec{e}_i\right) \\ &= \sum_{i=1}^n x_i \vec{f}(\vec{e}_i) \rightarrow (1) \end{aligned}$$

def: $\vec{f}(\vec{e}_i) = \vec{a}_i \in \mathbb{R}^m \rightarrow (2)$

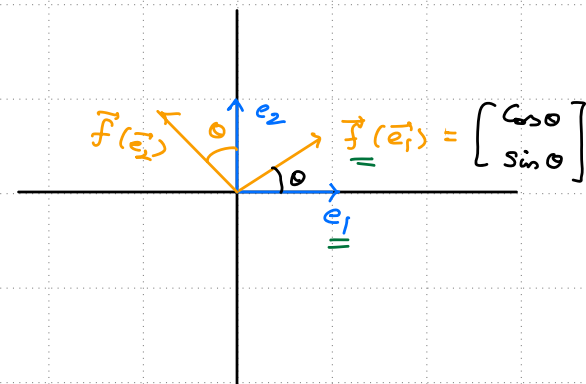
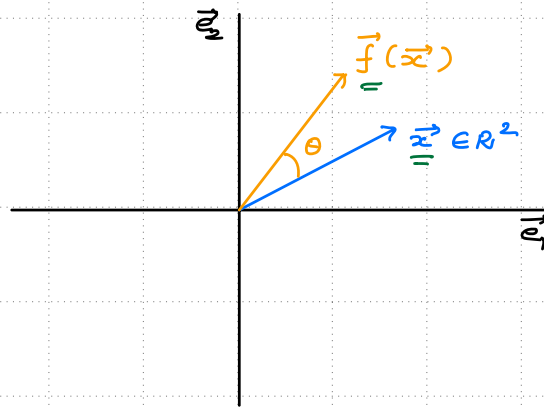
Sub (2) in (1) $\vec{f}(\vec{x}) = \sum_{i=1}^n x_i \vec{a}_i$
 $\searrow$ linear comb of vectors

$$= T\vec{x} \quad \text{where } T = [\vec{a}_1 \dots \vec{a}_n]$$

Obs 1: Every linear function $\vec{f}(\vec{x})$ can be written as $T\vec{x}$

Obs 2: every $A\vec{x}$ can be seen as a linear transformation (or) $\xrightarrow{\text{linear map}} \text{ from } \mathbb{R}^n \rightarrow \mathbb{R}^m$

Example: Rotation of a Vector in 2D



if $\vec{x}$ is rotated in C.C direction by an angle θ

compute corresponding T

$$T = [\vec{a}_1 \quad \vec{a}_2]$$

where $\vec{a}_1 = \vec{f}(\vec{e}_1)$ and

$$\vec{a}_2 = \vec{f}(\vec{e}_2)$$

$$\vec{a}_1 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

$$\vec{a}_2 = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

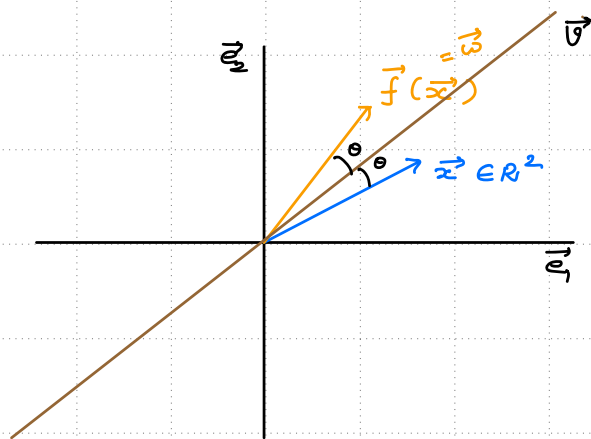
$$T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \leftarrow R_\theta$$

① does a $\vec{x} \in \mathbb{R}^n \neq \vec{0}$ exist s.t. $T\vec{x} = \vec{x}$ for every θ ? $\rightarrow$ No when $\vec{x} \in \mathbb{R}^2$

Related to
eigen vectors.

may be possible when $\vec{x} \in \mathbb{C}^2$

Reflection about a Line in 2D



See Handout for another version of
Transformation matrix

Reflection of a vector about $\vec{U}$?

$$T = [\vec{a}_1 \quad \vec{a}_2]$$

$$\vec{a}_1 = f(\vec{e}_1) \quad \& \quad \vec{a}_2 = f(\vec{e}_2)$$

Reflection of $\vec{x}$ around $\vec{U}$ is that vector (say $\vec{w}$)

$$a) \quad \text{Proj}_{\vec{U}} \vec{w} = \text{Proj}_{\vec{U}} \vec{x}$$

$$b) \quad \text{Proj}_{\vec{U}^\perp} \vec{w} = -\text{Proj}_{\vec{U}^\perp} \vec{x}$$

$$\vec{w} = \text{Proj}_{\vec{U}} \vec{x} - \text{Proj}_{\vec{U}^\perp} \vec{x}$$

$$= \text{Proj}_{\vec{U}} \vec{x} - (\vec{x} - \text{Proj}_{\vec{U}} \vec{x})$$

True only $\forall \vec{x} \in \mathbb{R}^2$
 $\downarrow$
 $[\vec{x} = \text{Proj}_{\vec{U}} \vec{x} + \text{Proj}_{\vec{U}^\perp} \vec{x}]$

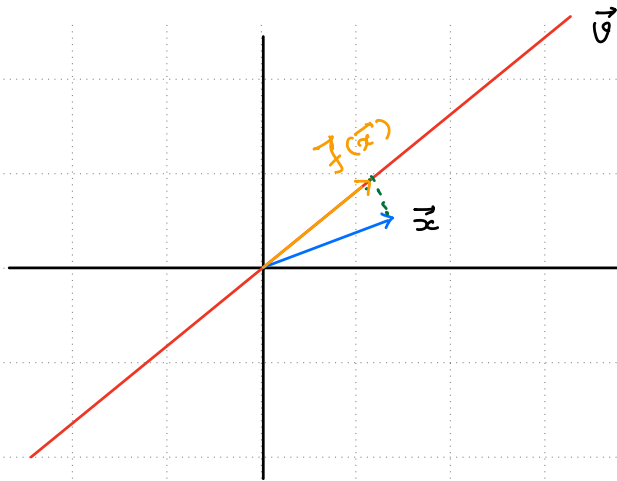
$$\vec{w} = 2 \text{Proj}_{\vec{U}} \vec{x} - \vec{x}$$

$$= 2 (\vec{x} \cdot \vec{e}_U) \vec{e}_U - \vec{x}$$

$$= 2 [\underbrace{\vec{e}_U \vec{e}_U^T}_{T.}] \vec{x} - \vec{x}$$

$T. \rightarrow$ House holder Transformation

Projection of a Vector onto a Line in 2D



$$f(x): \mathbb{R}^2 \rightarrow \mathbb{R}^2 = \text{Proj}_{\vec{v}} \vec{x}$$

Transformation matrix $T = [\vec{a}_1 \ \vec{a}_2]$

$$\begin{aligned} \vec{a}_1 = f(\vec{e}_1) &= \text{Proj}_{\vec{v}} \vec{e}_1 = (\vec{e}_1 \cdot \vec{e}_v) \vec{e}_v \\ &= (\vec{e}_v)_1 \vec{e}_v \end{aligned}$$

$$\begin{aligned} \vec{a}_2 = f(\vec{e}_2) &= \text{Proj}_{\vec{v}} \vec{e}_2 = (\vec{e}_2 \cdot \vec{e}_v) \vec{e}_v \\ &= (\vec{e}_v)_2 \vec{e}_v \end{aligned}$$

Example: $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

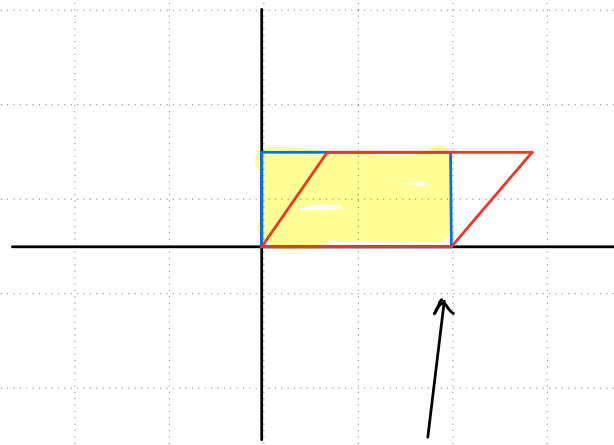
$$\vec{a}_1 = \text{Proj}_{\vec{v}} \vec{e}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = \begin{bmatrix} 1/5 \\ 2/5 \end{bmatrix}$$

$$\vec{a}_2 = \text{Proj}_{\vec{v}} \vec{e}_2 = \frac{2}{\sqrt{5}} \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = \begin{bmatrix} 2/5 \\ 4/5 \end{bmatrix}$$

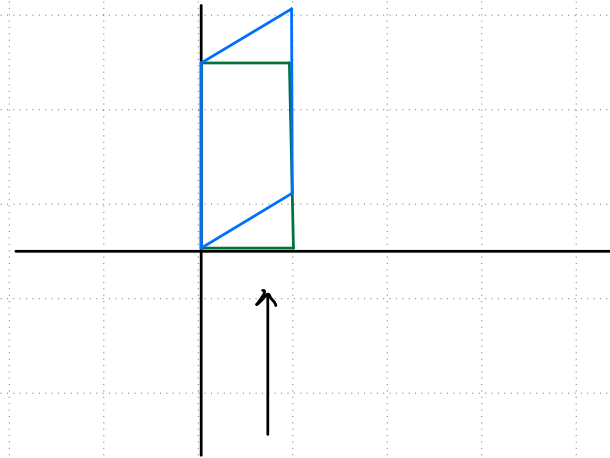
$$T = \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & 4/5 \end{bmatrix} \rightarrow \text{linearly dependent column vectors or row vectors}$$

Example: Shear Transformation in 2D

In general, any matrix can be considered as a linear Transformation



$$T_x = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix}$$



$$T_y = \begin{bmatrix} 1 & 0 \\ y & 1 \end{bmatrix}$$